

usepackage[a4paper,margin=0.65in]geometry

[745] (CONTINUED)

ON THE SCHWARZIAN DERIVATIVE AND THE POLYHEDRAL FUNCTIONS.

63. again P, Q given functions of z , and v a function of z determined by the equation

$$\frac{d^2v}{dz^2} + P \frac{dv}{dz} + Qv = 0;$$

and assume

$$y = wv.$$

Substituting this value of y in the first equation, we obtain for v an equation of the second order (the coefficients of which contain w), and we may make this identical with the second equation; viz. comparing the coefficients of the two equations, we thus have two equations each containing w ; and by eliminating w we obtain a differential equation of the third order between z and x . This is, in fact, the basis of Kummer's theory for the transformation of a hypergeometric series: the equation between z, x will be found presently in a different manner.

64. But if with Schwarz, instead of making the equation obtained for v as above identical with the given equation for v , we merely assume that the two equations are consistent, then there is nothing to determine the value of z , which may be regarded as an arbitrary function of x ; y and v are then functions of x , and w denotes the quotient $y \div v$ of these two functions, and as such satisfies an equation the form of which will depend on the assumed relation between z and x . In particular, if P and Q denote the same functions of z that p and q are of x ; and if we assume $z = x$, P, Q will become $= p, q$ respectively: the given equation in v will be

$$\frac{d^2v}{dx^2} + p \frac{dv}{dx} + qv = 0;$$

and w will thus denote the quotient of any two solutions of the equation

$$\frac{d^2y}{dx^2} + p \frac{dy}{dx} + qy = 0;$$

viz. writing $X = p^2 + 2\frac{dp}{dx} - 4q$, then, by what precedes, the equation for w will be

$$\{w, x\} = -\frac{1}{2}X.$$

65. Returning now to Kummer's problem, and considering y, v as solutions of the two differential equations respectively, we have $y = wv$, and taking any other two solutions denoted by these letters y_1, v_1 we have $y_1 = wv_1$, so that $\frac{y_1}{v_1} = \frac{y}{v}$: w is a function independent of the particular solutions of y, v ; calling the quotient of two solutions of the equation in v, s , also s_1 , denoting the quotient of two solutions of the equation in y , we have

$$\{s, x\} = -\frac{1}{2}Z, \quad \{s_1, z\} = -\frac{1}{2}Z,$$

and similarly $Z = P^2 + 2\frac{dP}{dz} - 4Q$, whence, writing as before $X = p^2 + 2\frac{dp}{dx} - 4q$, we have each of these equal quantities

we obtain

$$(\frac{dw}{dz})^2$$

where the accents denote differentiations in regard to an independent variable θ : viz. dividing by $2k'^2\lambda'^2$, this becomes

$$\{k, \theta\} + \frac{1}{2}k'^2 \left(\frac{1+k^2}{1-k^2} \right)^2 = \{\lambda, \theta\} + \frac{1}{2}\lambda'^2 \left(\frac{1+\lambda^2}{1-\lambda^2} \right)^2,$$

which is thus a particular case of Kummer's equation, k, λ corresponding to x, z respectively, and the values of X, Z being

$$X = - \left(\frac{1+k^2}{1-k^2} \right)^2, \quad Z = - \left(\frac{1+\lambda^2}{1-\lambda^2} \right)^2.$$

67. In the case of the hypergeometric series, the two differential equations of the second order are

$$\begin{aligned} \frac{d^2y}{dx^2} + \frac{\gamma - (\alpha + \beta + 1)x}{x(1-x)} \frac{dy}{dx} - \frac{\alpha\beta y}{x(1-x)} &= 0, \\ \frac{d^2v}{dz^2} + \frac{\gamma' - (\alpha' + \beta' + 1)z}{z(1-z)} \frac{dv}{dz} - \frac{\alpha'\beta' v}{z(1-z)} &= 0. \end{aligned}$$

Hence

$$p = \frac{\gamma + (1-x)(\gamma - \alpha - \beta - 1)x}{x(1-x)} = \frac{\gamma}{x} + \frac{\gamma - \alpha - \beta - 1}{1-x}, \quad q = -\frac{\alpha\beta}{x(1-x)},$$

and hence

$$X = p^2 + 2\frac{dp}{dx} - 4q = \frac{1-2\gamma}{x^2} + \frac{(\gamma - \alpha - \beta - 1)^2 + 2(\gamma - \alpha - \beta - 1) + 4\alpha\beta + 2\gamma(\gamma - \alpha - \beta - 1)}{(1-x)^2} + \frac{\text{etc}}{x(1-x)}$$

viz. writing

$$a = \frac{1}{4}(1 - \lambda^2), \quad b = \frac{1}{4}(1 - \mu^2), \quad c = \frac{1}{4}(1 - \nu^2),$$

where

$$\lambda^2 = (1 - \gamma)^2, \quad \mu^2 = (\gamma - \alpha - \beta)^2, \quad \nu^2 = (\alpha - \beta)^2,$$

and putting in the formula $x - 1, = -(1 - x)$, we have

$$-\frac{1}{4} \left(p^2 + 2\frac{dp}{dx} - 4q \right) = \frac{a}{x^2} + \frac{b}{(x-1)^2} + \frac{-a-b+c}{x(x-1)} = \frac{a}{x^2} + \frac{b}{(x-1)^2} + \frac{c-a-b}{x \cdot x-1} = (a, b, c \ x, x-1);$$

with a like formula for $\frac{1}{4} \left(P^2 + 2\frac{dP}{dz} - 4Q \right)$. We then have

$$y = wv, \quad v = x^{c-1}(1-x)^{\gamma-\alpha-\beta-1} z^{c'}(1-z)^{-\gamma'+\alpha'+\beta'+1} \dots,$$

and the differential equation of the third order for the determination of z is

$$\{z, x\} = (a_1, b_1, c_1 \mid z, z-1) \left(\frac{dz}{dx} \right)^2 - (a, b, c \mid x, x-1) = 0,$$

where a_1, b_1, c_1 are the same functions of α', β', γ' which a, b, c are of α, β, γ . This is, in effect, Kummer's equation for the transformation of the hypergeometric series.

68. And in like manner the Schwarzian equation for the determination of z , the quotient of two solutions, is

$$\{s, x\} = (a, b, c \mid x, x-1).$$

PART II. THE POLYHEDRAL FUNCTIONS.

Origin and Properties. Art. Nos. 69 to 80.

69. The geometrical functions in question are connected with the geometrical forms (these being regarded as situate on a spherical surface), and with the stereographic projections of these figures:—

1. Polygon or
2. Double Pyramid,*
3. Tetrahedron,
4. Octahedron and Cube,
5. Dodecahedron and Icosahedron,

* Prof. Klein regards 1 as belonging to the polygon and 2 to the double pyramid: it seems to me that the fundamental figure, to which 1 and 2 each of them belong, is the polygon.

Consider a spherical surface and upon it any number of points: take at pleasure any point as South Pole, this determines the plane of the equator; and the stereographic projection of any point is the intersection with the plane of the equator of the line joining the point with the South Pole. To fix the ideas take the radius of the sphere as unity: let the axes of x and y be drawn in the plane of the equator in longitudes 0° and 90° respectively, and the axis of z upwards through the North Pole: the position of a point on the sphere is determined by means of its N.P.D. θ and longitude f : moreover we take X, Y, Z for the coordinates of the point on the surface, and x, y for those of its projection; and we then have

$$X, Y, Z = \sin \theta \cos f, \sin \theta \sin f, \cos \theta;$$

$$x = \frac{X}{1+Z} = \tan \frac{1}{2}\theta \cos f, \quad y = \frac{Y}{1+Z} = \tan \frac{1}{2}\theta \sin f,$$

and conversely

$$X, Y, Z = 2x, 2y, 1 - x^2 - y^2 \div (1 + x^2 + y^2).$$

We represent the point (X, Y, Z) on the spherical surface by means of the magnitude $x + iy$, $= \tan \frac{1}{2}\theta (\cos f + i \sin f)$, or say by the linear factor $s - (x + iy)$: and similarly any system of points on the surface by means of the system of magnitudes $x + iy$, or say by the function $\prod[s - (x + iy)]$, denoting in this manner the product of the linear factors which correspond to the different points respectively.

70. It will presently appear that, if (considering different stereographic projections, that is, a different position of the South Pole) we take x', y' as the coordinates of the new projection of the point, then $x' + iy'$ is a homographic function of $x + iy$: and consequently that the functions of s , which belong to the different projections, are linear transformations one of the other: but at present we consider a single projection.

It may be proper to remark that the figures in question are spherical figures having for summits points on the spherical surface, for sides arcs of great circle joining two summits, and faces which are portions of the spherical surface: the mid-points of the sides (or edges) and the centres of the faces are of course points on the spherical surface.

71. (1), (2). Considering a regular polygon formed by n summits on the equator, the longitude of one of them being 0° , then the stereographic projections correspond with the points themselves, and the values of $x + iy$ are

$$1, \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}, \dots, \cos \frac{(n-1)2\pi}{n} + i \sin \frac{(n-1)2\pi}{n}.$$

The corresponding function of s is $s^n - 1$.

The values of $x + iy$ for the mid-points of the sides are

$$\cos \frac{\pi}{n} + i \sin \frac{\pi}{n}, \cos \frac{3\pi}{n} + i \sin \frac{3\pi}{n}, \dots, \cos \frac{(2n-1)\pi}{n} + i \sin \frac{(2n-1)\pi}{n}.$$

The corresponding function of s is $s^n + 1$.

The North and South Poles, which form with the n points a double pyramid of $n + 2$ summits, correspond to the values $s = 0$ and $s = \infty$. We have thus

$$s(1 - s^n)(s^n - 1)$$

as the function corresponding to the double pyramid.

72. (3). Considering for a moment the tetrahedron as a figure with rectilinear edges, this is so placed that two opposite edges are horizontal, and that the vertical planes passing through the centre and the upper and lower edges respectively are the meridian planes; the upper edge has the longitudes $\pm 45^\circ$ and the lower edge the longitudes $135^\circ, 225^\circ$. We thus explain the position of the spherical figure.

Corresponding to the summits we have the function $s^4 - 2i\sqrt{3}s^2 + 1$. In fact, the equation $s^4 - 2i\sqrt{3}s^2 + 1 = 0$ gives $s^2 = i(\sqrt{3} \pm 2)$, and hence the values of s are the four values of $x + iy$ shown in the annexed table for the values of X, Y, Z , and $x + iy$ for the summits of the tetrahedron.

long.	45°	135°	225°	315°
X	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$
Y	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$
Z	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$
$x + iy$	$\frac{1+i}{\sqrt{3}-1}$	$\frac{-1+i}{\sqrt{3}+1}$	$\frac{-1-i}{\sqrt{3}-1}$	$\frac{1-i}{\sqrt{3}+1}$

Corresponding to the centres of the faces, or summits of the opposite tetrahedron, we have the function $s^4 + 2i\sqrt{3}s^2 + 1$.

Corresponding to the mid-points of the sides, we have the function

$$s(1-s^4)(s^4-1);$$

viz. the points in question are the four points $s = \pm 1$, $s = \pm i$, the North Pole $s = 0$, the South Pole $s = \infty$, and the four points on the equator at longitudes 0° , 90° , 180° , 270° .

73. (4). The octahedron is placed with two of its summits as poles, and the other four summits in the equator at longitudes 0° , 90° , 180° , 270° respectively: the values of s are, as in the last case, 0 , ∞ , ± 1 , $\pm i$, and the function is

$$s(1-s^4)(s^4-1).$$

The function for the centres of the faces, or summits of the cube, is $s^8 + 14s^4 + 1$.

The function for the mid-points of the sides of the octahedron or of the cube is

$$s^{12} - 33s^8 - 33s^4 + 1.$$

74. (5). The Icosahedron is placed with two of its summits for poles; five summits lying in a small circle above the plane of the equator at longitudes 0° , 72° , 144° , 216° , 288° ; and the remaining five summits in the corresponding small circle below the equator at longitudes 36° , 108° , 180° , 252° and 324° .

The function for the summits of the Icosahedron is

$$s(1-s^{10})(s^{10}+11s^5-1).$$

The function for the centres of the faces of the Icosahedron, or summits of the Dodecahedron, is

$$s^{20} - 228s^{15} + 494s^{10} + 228s^5 - 1.$$

The function for the mid-points of the sides of the Icosahedron or the Dodecahedron is

$$s^{30} - 522s^{25} + 10005s^{20} + 0s^{15} - 10005s^{10} + 522s^5 + 1.$$

I give for the present these results without demonstration.

75. Writing $\frac{x}{y}$ for the s , so as to obtain from the foregoing values $x + iy$ of these homogeneous functions ($; x, y$)^{*n*},—it will be recollected that the x, y have nothing to do with the x, y of § 69—the forms which have thus presented themselves may be denoted as follows:

$$\begin{aligned} (3) : \quad f3 &= (1, -2i\sqrt{3}, 1; x, y)^4, \\ h3 &= (f3, f3)^2, \quad t3 = xy(x^4 - y^4), \\ (4) : \quad f4 &= xy(x^4 - y^4), \\ h4 &= (1, 14, 1; x^4, y^4)^2, \\ t4 &= (1, -33, -33, 1; x^4, y^4)^3, \\ (5) : \quad f5 &= xy(1, 11, -1; x^5, y^5)^2, \\ h5 &= (1, -228, +494, +228, -1; x^5, y^5)^4, \\ t5 &= (1, -522, 10005, 0, -10005, 522, 1; x^5, y^5)^6; \end{aligned}$$

where observe that $f4$ is the same function as $t3$. In each set of functions f, h, t , we have h and t covariants of f , viz. disregarding numerical factors, h is the Hessian, or derivative $(f, f)^2$, and t is the derivative (f, h) .

76. Since $f4$ is the same function as $t3$, we have of course $f4, h4$ and $t4$ themselves covariants of $f3$; but it is convenient to separate the two systems.

77. It is to be observed that $f3$ is a quartic function having its quadrinvariant $(f) = 0$; but independently of this, $f3$, qua quartic function, has only the covariants $h3$ and $t3$ (the Hessian and the cubicovariant respectively), viz. every other covariant is a rational and integral function of $f3, h3$ and $t3$. In particular, $h4$ and $t4$ are rational and integral functions of $f3, h3$ and $t3$; but inasmuch as $f3$ and $h3$ are not covariants of $f4$, this is not a property of $h4$ and $t4$ considered as covariants of $f4$, and the relation in question need not be attended to.

78. It has been just stated that $f3$ qua quartic function has only the covariants $h3$ and $t3$: $f4$ qua special sextic function (in the sense explained) and $f5$ qua dodecadic function have the like property, viz. $f4$ has only the covariants $h4$ and $t4$; $f5$ only the covariants $h5$ and $t5$. Hence $f3, f4, f5$ are “Prime-forms” in the sense defined in the paper by Fuchs, of 1875, and itself, viz. a Prime-form has no covariant of a lower order than itself, and also no covariant of a higher order which is a power of a form of a lower order.

79. The same functions have also the property that they are functions transformable into themselves by means of a group of linear transformations, and in this point of view they were considered in the nearly contemporaneous paper by Klein, of 1875; it is in this paper shown that the functions so transformable into themselves must be Polyhedral functions as above, the linear transformations in fact corresponding to the rotations whereby the spherical polyhedron can be brought into coincidence with its own original position. This theory will be presently given.

80. It is to be observed that, if U, V are functions $(; x, y)^n$ of the order n , then using the accent to denote differentiation in regard to x , $UV' - U'V$ and (U, V) differ only by a numerical factor: and further that, writing as before $s = \frac{x}{y}$, and in the expression $UV' - U'V$ regarding U, V as functions $(; s, 1)^n$, and the accent as denoting differentiation in regard to s , we have $UV' - U'V$ and (U, V) differing by a numerical factor only. We have in the PQR -Table, the lines 3, 4, 5, P, Q, R equal to given numerical multiples of $h^\alpha, t^\beta, f^\gamma$, the indices α, β, γ being such as to make these to be functions of the same degree: hence, neglecting numerical multipliers,

$$PQ' - P'Q = h^{\alpha-1}t^{\beta-1}(h, t),$$

and the theorem that $PQ' - P'Q$ is equal to a function $(; f^\delta)$, which contains only factors of R , is in fact the theorem that $(h, t), (h, f)$, and (t, f) are each of them equal to a term or product of f, h, t ; which is a result included in the theorem that f has only the covariants h and t . And by this last theorem we know already how from R , assumed to be known, we can derive P and Q : viz. R is a power of f ; and we thence have $h = (f, f)^2$ and $t = (h, f)$, equations giving the functions h and t , upon which P, Q depend.

Covariantive Formulæ. Art. Nos. 81 to 84.

81. The various covariantive formulæ will be given with their proper numerical coefficients.

Tetrahedron function. f, h, t stand for the before-mentioned values, f^3, h^3, t^3

$$(P, Q, R = h^3, -12i\sqrt{3}t^2, -f^6).$$

For f^3 .

$$\begin{aligned} (a, b, c, d, e) &= (1, 0, -\frac{1}{3}i\sqrt{3}, 0, 1), \\ \frac{1}{4}(f, f)^2 &= -\frac{1}{2}i\sqrt{3}h, \quad \frac{1}{4}(f, f)^4 = \frac{4}{3}, \quad \frac{1}{4}(f, f)^6 = 0, \\ (f, h) &= -32i\sqrt{3}t, \quad \frac{1}{4}(f, h)^2 = 1152f = 1152\frac{4}{3}f^2, \\ (h, t) &= 96i\sqrt{3}f, \quad \frac{1}{4}(t, t)^2 = -8fh^2, \end{aligned}$$

$$\begin{aligned}
(f, t) &= -96i\sqrt{3}h^2, \\
h^3 - f^3 - 12i\sqrt{3}t^2 &= 0, \\
fh &= (1, 14, 1; x^4, y^4)^2 (= f4).
\end{aligned}$$

It is convenient to remark that t^2, f^3, h^3 being of the same order, we have

$$P(f, h) + Q(f, t)^2 + R(h, t)^2 = 0,$$

that is,

$$t^2 \cdot 3f^2(f, h) + f^3 \cdot 2h(h, t) + h^3 \cdot 2 \cdot 3t(t, f) = 0,$$

an equation which, substituting the values for $(f, h), (h, t), (t, f)$ their values, reduces itself to $h^3 - f^3 - 12i\sqrt{3}t^2 = 0$; and we have thus a verification of the before-mentioned relation and the values of $(f, h), (h, t)$ and (t, f) . The like remark applies to the other two cases, which follow.

82. *Hexahedron function.* f, h, t stand for the before-mentioned values

$$f4, h4, t4 \quad (P, Q, R = h^3, -t^2, -108f^4).$$

For $f4$.

$$\begin{aligned}
(a, b, c, d, e, f, g) &= (0, \frac{1}{6}, 0, 0, 0, -\frac{1}{6}, 0), \\
\frac{1}{4}(f, f)^2 &= -25h, \quad \frac{1}{4}(f, f)^4 = 0, \quad \frac{1}{4}(f, f)^6 = (720)^2 \frac{1}{6}, \\
(f, h) &= -8t, \quad \frac{1}{4}(f, h)^2 = 3 \cdot 2^8 \cdot 7^2 \cdot h^2, \\
(f, t) &= -12h^2, \quad \frac{1}{4}(t, t)^2 = 2^4 \cdot 3^2 \cdot 17^2 \cdot f^2, \\
(h, t) &= -1728f^3, \\
h^3 - t^2 - 108f^4 &= 0.
\end{aligned}$$

83. *Dodecahedron function.* f, h, t stand for the before-mentioned values

$$f5, h5, t5 \quad (P, Q, R = h^3, -t^2, -1728f^5).$$

For $f5$.

$$\begin{aligned}
(a, b, c, d, e, f, g, h, i, j, k, l, m) &= (0, \frac{1}{12}, 0, 0, 0, 0, \frac{11}{12}, 0, 0, 0, 0, -\frac{1}{12}, 0), \\
\frac{1}{4}(f, f)^2 &= -121h, \quad \frac{1}{4}(f, f)^4 = 0, \quad \frac{1}{4}(f, f)^6 = \frac{1}{4}(924)^2(720)^2 \frac{11}{12}f^2, \\
\frac{1}{4}(f, f)^8 &= \frac{1}{4}(924)^2(720)^2 \frac{1}{12}f^2, \quad \frac{1}{4}(f, f)^{10} = 0, \quad \frac{1}{4}(f, f)^{12} = 0, \\
(f, h) &= -20t, \quad \frac{1}{4}(f, h)^2 = 173280h^2, \\
(f, t) &= -30h^2, \quad \frac{1}{4}(t, t)^2 = 9082800f^2h, \\
(h, t) &= -86400f^4, \\
h^3 - t^2 - 1728f^5 &= 0.
\end{aligned}$$

84. We have

$$t = (x^5 + y^5) (1, 522, -10005, -522, 1; x^5, y^5)^4.$$

Write

$$\xi = (x^5 + y^5) (1, 2, 6, -2, 1; x^5, y^5)^4,$$

then

$$t = \xi (1, -10, 45; x, y)^2.$$

Or putting

$$\rho = \frac{1}{xy} \sqrt[4]{(x^5 + y^5) (1, 2, 6, -2, 1; x^5, y^5)^4},$$

that is,

$$\rho = \frac{1}{xy} \sqrt[4]{(x^{10} + 11x^5y^5 - y^{10}) (\dots)},$$

Writing $k = P/f$, then

$$\rho^3 - 10\rho^2 + 45\rho = \frac{k}{f}. \quad (\text{Klein.})$$

Investigation of the forms $f5$ and $h5$. Art. Nos. 85 and 86.

85. Writing for shortness¹ $k = \tan \alpha = \sqrt{\frac{5 - \sqrt{5}}{10}}$, and $g = \cos 36^\circ + i \sin 36^\circ$, then the values of $x + iy$ corresponding to the summits of the Icosahedron are

$$0, k, kg, kg^2, kg^3, kg^4, k^{-1}, k^{-1}g, k^{-1}g^2, k^{-1}g^3, k^{-1}g^4, \infty,$$

and the function $f5$ is thus

$$= s (1 - s^5/k^5)(s^5 - k^{-5}),$$

where the product of the last two factors is $s^{10} + (k^{-5} - k^5)s^5 - 1$. We have

$$k^{-5} - k^5 = \frac{1}{10} \sqrt{5} (5\sqrt{5} - 11), \quad = \frac{1}{2} (\sqrt{5} - 11) \cdot \sqrt{5}/5$$

and consequently $k^{-5} - k^5 = 11$; or the function is

$$s (1 - s^{10})(s^{10} + 11s^5 - 1).$$

¹ k is the α and h the β of the Table, No. 99.

86. Similarly, writing for shortness²

$$\cos \frac{\gamma}{2} = \frac{1}{2} \sqrt{\frac{5 + \sqrt{5}}{2\sqrt{5}}}, \quad \cos^2 \frac{\gamma}{2} = \frac{5 + 2\sqrt{5}}{15}, \quad \sin^2 \frac{\gamma}{2} = \frac{10 - 2\sqrt{5}}{15},$$

and

$$l = \tan \frac{1}{2}\gamma, \quad l' = \tan \frac{1}{2}\gamma', \quad \text{where } \cos \frac{\gamma'}{2} = \frac{3 + \sqrt{5}}{\dots}, \quad \sin \frac{\gamma'}{2} = \frac{3 - \sqrt{5}}{\dots},$$

and therefore

$$l^5 = \frac{\cos \gamma}{\sin \gamma};$$

$g = \cos 36^\circ + i \sin 36^\circ$ as before, then the values of $x + iy$ for the summits of the dodecahedron are

$$l g, l g^2, l g^3, l g^4, l g^5; \quad l' g, l' g^2, l' g^3, l' g^4, l' g^5; \\ l^{-1}, l^{-1} g, l^{-1} g^2, l^{-1} g^3, l^{-1} g^4; \quad l'^{-1}, l'^{-1} g, l'^{-1} g^2, l'^{-1} g^3, l'^{-1} g^4.$$

The function $h5$ is therefore

$$= s^{20} + s^{15}(l^5 - l^{-5}) + l^5 \cdot l'^5 s^{10} + s^5(l'^5 - l'^{-5}) - 1.$$

We have

$$l^{-5} - l^5 = \frac{2 \cos \gamma}{\sin^5 \gamma} = \frac{(1 - \cos \gamma)^5}{\sin^5 \gamma} \cdot \frac{1}{(1 + \cos \gamma)^5} = \frac{(5 + 10 \cos^2 \gamma + \cos^4 \gamma)}{\sin^5 \gamma} = \frac{384 + 128 \cos \gamma}{45 \sin^5 \gamma} = \frac{64\sqrt{5}}{45} (6$$

viz. this last identity depends on

$$\frac{2}{45}(3 + \sqrt{5})(6 + \sqrt{5}) = (114 + 50\sqrt{5}) \sin^4 \gamma,$$

that is,

$$160(3 + \sqrt{5})(6 + \sqrt{5}) = (114 + 50\sqrt{5})(120 - 40\sqrt{5}),$$

or

$$2(3 + \sqrt{5})(6 + \sqrt{5}) = (57 + 25\sqrt{5})(3 - \sqrt{5}),$$

or finally

$$(7 + 3\sqrt{5})(6 + \sqrt{5}) = 57 + 25\sqrt{5},$$

which is right.

Similarly

$$l'^{-5} - l'^5 = 114 - 50\sqrt{5},$$

and observing that the sum and product of $114 + 50\sqrt{5}$, $114 - 50\sqrt{5}$ are $= 228$ and 496 respectively, the required function of s is

$$= s^{20} - 1 - 228(s^{15} - s^5) + 496 s^{10},$$

² α is the α , γ is the γ , and γ' the $\alpha - \beta$ of the Table, No. 99.

$$= s^{20} - 228 s^{15} + 494 s^{10} + 228 s^5 + 1,$$

$$(= h5)$$

which is the required value of $h5$.

Invariantive property of the Stereographic Projection.

Art. Nos. 87 to 93.

87. The before-mentioned theorem that the functions derived from two different stereographic projections of the same point are linear transformations one of the other, may be thus stated:—

Considering on the surface of a sphere, two fixed points A and B ; and determining the position of a point C , first in regard to A by its distance θ and azimuth f , and next in regard to B by its distance θ' and azimuth f' , the azimuths from the great circle ABx which joins the two points A and B , then we have

$$s = \tan \frac{1}{2}\theta (\cos f + i \sin f), \quad s' = \tan \frac{1}{2}\theta' (\cos f' + i \sin f'),$$

calling them homographic functions one of the other; and putting the distance $AB = c$, the relation between them in fact is

$$s' = \frac{s - \tan \frac{1}{2}c}{1 + s \tan \frac{1}{2}c};$$

or, what is the same thing, observing that

$$s s' = \tan \frac{1}{2}\theta \tan \frac{1}{2}\theta' \{\cos(f + f') + i \sin(f + f')\},$$

we have the two equations

$$\tan \frac{1}{2}c \{1 + \tan \frac{1}{2}\theta \tan \frac{1}{2}\theta' \cos(f + f')\} = \tan \frac{1}{2}\theta \cos f - \tan \frac{1}{2}\theta' \cos f',$$

$$\tan \frac{1}{2}c \{1 + \tan \frac{1}{2}\theta \tan \frac{1}{2}\theta' \sin(f + f')\} = \tan \frac{1}{2}\theta \sin f - \tan \frac{1}{2}\theta' \sin f'.$$

88. If we denote the angles of the spherical triangle C, A, B , and the opposite sides by c (as before), a, b , then $\theta, \theta' = b, a$; $f, f' = A, \pi - B$, whence

$$s = \tan \frac{1}{2}b (\cos A + i \sin A), \quad s' = -\tan \frac{1}{2}a (\cos B - i \sin B),$$

and we have between the spherical sides a, b, c and angles A, B of the spherical triangle the relations

$$\tan \frac{1}{2}c \{1 - \tan \frac{1}{2}a \tan \frac{1}{2}b \cos(A - B)\} = \tan \frac{1}{2}b \cos A + \tan \frac{1}{2}a \cos B,$$

$$\tan \frac{1}{2}c \{1 - \tan \frac{1}{2}a \tan \frac{1}{2}b \sin(A - B)\} = \tan \frac{1}{2}b \sin A - \tan \frac{1}{2}a \sin B;$$

which equations may be verified by means of the ordinary formulae of Spherical Trigonometry.

89. But it is interesting to give the proof with rectangular coordinates. Taking (X, Y, Z) , (X_1, Y_1, Z_1) for the coordinates, referred to two different sets of axes, of a point on the spherical surface: also x, y, x_1, y_1 for the coordinates of the corresponding stereographic projections, we have

$$(X_1 : Y_1 : Z_1) = (\alpha, \beta, \gamma; \alpha', \beta', \gamma'; \alpha'', \beta'', \gamma'')(X : Y : Z),$$

$$X : Y : Z : 1 = 2x : 2y : 1 - x^2 - y^2 : 1 + x^2 + y^2,$$

$$X_1 : Y_1 : Z_1 : 1 = 2x_1 : 2y_1 : 1 - x_1^2 - y_1^2 : 1 + x_1^2 + y_1^2,$$

and thence

$$x_1 : y_1 : 1 = 2\alpha x + 2\beta y + \gamma(1 - x^2 - y^2) : 2\alpha'x + 2\beta'y + \gamma'(1 - x^2 - y^2) : 1 + x^2 + y^2 + 2\alpha''x + 2\beta''y + \gamma''(1 - x^2 - y^2),$$

90. Introducing z, z_1 for homogeneity, or writing $\frac{x}{z}, \frac{y}{z}$ and $\frac{x_1}{z_1}, \frac{y_1}{z_1}$ in place of x, y and x_1, y_1 , respectively, we have

$$x_1 = 2\alpha x + 2\beta y + \gamma(z^2 - x^2 - y^2) = (-\gamma, -\gamma, +\gamma, \beta, \alpha, 0; x, y, z)^2,$$

$$y_1 = 2\alpha'x + 2\beta'y + \gamma'(z^2 - x^2 - y^2) = (-\gamma', -\gamma', +\gamma', \beta', \alpha', 0; x, y, z)^2,$$

$$z_1 = z^2 + x^2 + y^2 + 2\alpha''x + 2\beta''y + \gamma''(z^2 - x^2 - y^2) = (1 - \gamma'', 1 - \gamma'', 1 + \gamma'', \beta'', \alpha'', 0; x, y, z)^2,$$

and thence without difficulty

$$x_1 + iy_1 = \frac{1}{\gamma + \gamma''} \{(1 + \gamma'')z + (\alpha'' + i\beta'')(x - iy)\} \{(1 - \gamma'')z + (-\alpha'' + i\beta'')(x + iy)\},$$

$$z_1 = \frac{1}{\gamma + \gamma''} \{(1 - \gamma'')z - (\alpha'' + i\beta'')(x - iy)\} \{(1 + \gamma'')z + (\alpha'' - i\beta'')(x + iy)\},$$

$$x_1 - iy_1 = MN : NL : LM \quad (L, M, N \text{ linear functions of } z, x + iy, x - iy)$$

showing that the relation between two stereographic projections of the same spherical figure is in fact that of a quadric transformation, the fundamental points in each figure being an arbitrary point and the two circular points at infinity: or, what is the same thing, to any line in the one figure there corresponds a circle in the other figure, which is the “circular relation” of Möbius.

91. The actual values are

$$\frac{x_1 + iy_1}{z_1} = \frac{1 + \gamma''}{\gamma + \gamma''} \cdot \frac{(1 - \gamma'')z - (\alpha'' - i\beta'')(x + iy)}{(1 + \gamma'')z + (\alpha'' - i\beta'')(x + iy)},$$

$$\frac{x_1 - iy_1}{z_1} = \frac{1 + \gamma''}{\gamma - \gamma''} \cdot \frac{(1 - \gamma'')z - (\alpha'' + i\beta'')(x - iy)}{(1 + \gamma'')z + (\alpha'' + i\beta'')(x - iy)},$$

viz. attending only to the former of these, we have $\frac{x_1 + iy_1}{z_1}$ a homographic function of $\frac{x + iy}{z}$, which is the before-mentioned theorem.

92. Supposing that the transformation from (X, Y, Z) to (X_1, Y_1, Z_1) is made by a rotation, the coordinates of which are λ, μ, ν , that is, if f, g, h are the inclinations of the resultant axis to the axes of x, y, z respectively, and θ the angle of rotation, putting $\lambda, \mu, \nu = \tan \frac{1}{2}\theta \cos f, \tan \frac{1}{2}\theta \cos g, \tan \frac{1}{2}\theta \cos h$: then the coefficients of transformation are

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix} = \frac{1}{1 + \lambda^2 + \mu^2 + \nu^2} \begin{pmatrix} 1 + \lambda^2 - \mu^2 - \nu^2 & 2(\lambda\mu - \nu) & 2(\lambda\nu + \mu) \\ 2(\lambda\mu + \nu) & 1 - \lambda^2 + \mu^2 - \nu^2 & 2(\mu\nu - \lambda) \\ 2(\lambda\nu - \mu) & 2(\mu\nu + \lambda) & 1 - \lambda^2 - \mu^2 + \nu^2 \end{pmatrix}.$$

Substituting these values, the formulae become, after an easy reduction,

$$\frac{x_1 + iy_1}{z_1} = \frac{-(\nu + i)(x + iy) + (\lambda + i\mu)z}{(\lambda - i\mu)(x + iy) + (\nu - i)z},$$

$$\frac{x_1 - iy_1}{z_1} = \frac{-(\nu - i)(x - iy) + (\lambda - i\mu)z}{(\lambda + i\mu)(x - iy) + (\nu + i)z},$$

attending to the former of these, and writing for greater simplicity s, s_1 respectively, we have

$$s_1 = \frac{-(\nu + i)s + (\lambda + i\mu)}{(\lambda - i\mu)s + (\nu - i)},$$

or writing this

$$s_1 = \frac{As + B}{Cs + D},$$

then

$$A : B : C : D = -\nu - i : \lambda + i\mu : \lambda - i\mu : \nu - i.$$

93. I call to mind that the condition, in order that the homographic transformation $s_1 = (As + B) \div (Cs + D)$ may be periodic of the order n , is

$$(A + D)^2 - 4(AD - BC) \cos^2 \frac{m\pi}{n} = 0,$$

m being an integer different from zero and prime to n . In particular, when $n = 2$, it is $A + D = 0$: $n = 3$, it is $A^2 + AD + D^2 + BC = 0$: $n = 4$, it is $A^2 + D^2 + 2BC = 0$: and $n = 5$, it is $(A + D)^2 - \frac{1}{2}(3 \pm \sqrt{5})(AD - BC) = 0$.

Groups of homographic transformations. Art. Nos. 94 and 95.

94. The formulae just obtained serve to connect the theory of the rotations of a polyhedron with that of the homographic transformations s into $(As + B) \div (Cs + D)$; and, corresponding to the rotations which leave the polyhedron unaltered, we have groups of homographic transformations. We have thus, corresponding to the cases of the tetrahedron, the cube and the octahedron, and the dodecahedron and icosahedron respectively, groups of 12, of 24, and of 60 homographic transformations s into

$(As + B) \div (Cs + D)$. The group of 60 and the group of 24 include each of them as part of itself the group of 12; it is further to be remarked that the group of 12 may be regarded as that of the positive substitutions upon the four letters $abcd$, the group of 24 as that of all the substitutions upon the four letters, and the group of 60 as that of the positive substitutions upon the five letters $abcde$.

95. I call to mind that a group of functional symbols $1, \alpha, \beta, \dots$ can always be expressed in the equivalent form $1, \phi\alpha\phi^{-1}, \phi\beta\phi^{-1}, \dots$, where ϕ is any functional symbol whatever: clearly, $\alpha, \beta, \dots$ being homographic transformations, then, ϕ being any homographic transformation whatever, the new symbols $\phi\alpha\phi^{-1}, \phi\beta\phi^{-1}, \dots$ will also be homographic transformations; and thus the group of homographic transformations can be expressed in various equivalent forms: these correspond to the different positions of the polyhedron in regard to the axes of coordinates; viz. in the three cases which it is proper to consider, the dodecahedron, we may have the Pole at a point Θ , that is, in the language of art. Nos. as presently explained, the axis of z passing through the midpoint of a side, 1°: through a summit; that is, the Pole at a point A , 2°: and there are different forms which correspond to a Pole at a point B , 3°.

The regular Polyhedra. Art. Nos. 96 to 103.

96. We require a theory of the regular Polyhedra considered as systems of points on a sphere. I refer to my two papers [375] and [679]. In the latter paper, I remark that, considering the five regular figures drawn in proper relation to each other on the same spherical surface, the only points which have to be considered are 12 points A , 20 points B , 30 points Θ , and 60 points Φ . Describing these by reference to the dodecahedron, the points A are the centres of the faces, the points B are the summits, the points Θ are the midpoints of the sides, and the points Φ are the midpoints of the diagonals of the faces, each point Φ being given as the intersection of a face of the dodecahedron and a side of the icosahedron, which there intersect at right angles. Or describing them by reference to the icosahedron, the points A are the summits, the points B are the centres of the faces, the points Θ are the midpoints of the sides: and the points Φ are points lying by threes on the faces of the icosahedron, which there is no occasion to attend to; viz.

Φ is the point common to the face and a line BB joining the centres of two adjacent faces at right angles.

97. The points Φ are comparatively unimportant, and it is proper first to attend only to the 12 points A , the 20 points B , and the 30 points Θ : these form 6 pairs of opposite points A , 10 pairs of opposite points B , and 15 pairs of opposite points Θ . Considering the diameters through each pair of opposite points Θ , we have thus a system of 15 axes, which in fact form 5 sets each of 3 rectangular axes: attending to any one of such sets, the diametral plane at right angles to one of the three axes contains of course the other two axes: it contains also two axes each through a pair of opposite points A , and two axes each through a pair of opposite points B . If instead of the plane we consider its intersection with the sphere, we have thus on the sphere 15 circles each containing 4 points Θ ,

4 points A and 4 points B . The fifteen circles intersect by fives in the pairs of opposite points A , by threes in the pairs of opposite points B , and by twos in the pairs of opposite points Θ ; the mutual inclinations of successive circles at the points A , B , Θ being $= 36^\circ$, 60° and 90° respectively. The whole number $15 \cdot 14 = 210$, of the intersections of the two circles two and two together is thus made up of the 12 points A each counting 10 times, the 20 points B each counting 3 times, and the 30 points Θ each counting once; $210 = 120 + 60 + 30$.

98. The angular magnitudes which present themselves are all obtained from the dodecahedral pentagon, as shown in the annexed figure, in which the angle subtended by a side at the centre is $= 72^\circ$, and the angle between two adjacent sides is $= 120^\circ$.

[Figure: dodecahedral pentagon with vertices $B_1, B_2, \dots, B_5$, centre A , midpoints $\Theta, \Theta', \dots$.]

We write $A\Theta = \alpha$, $B\Theta = \beta$, $AB = \gamma$, $B_1B_2 = x$, $\angle B_1B_2B = \theta$, $\Theta B_1 = g$, $\angle \Theta B_1B = \phi$.

From the triangle $A\Theta B$, the angles of which are 36° , 90° , 60° and the opposite β , γ , α , we find the values of α , β , γ , and these are such that $\alpha + \beta + \gamma = \frac{1}{2}\pi$.

From the triangle B_2BB_1 , where the sides B_1B , BB_2 and the included angle are 2β , 2β , 120° , we have the opposite side x , and the other two angles each $= \theta$.

From the triangle $B_1\Theta B_2$, where the sides B_1B , $B_2\Theta$ and the included angle $BB_2\Theta$ are 2β , β , 120° , we find the opposite side g , and the angle $B_2\Theta B_1 = 45^\circ$.

Hence each of the angles $B_1\Theta B$, $B_2\Theta B_1$, being $= 45^\circ$, the angle $B_1\Theta B_2$ is $= 90^\circ$: in this triangle the hypotenuse B_1B_2 is $= x$, and the other two sides are $= g$: whence we have $\cos x = \cos^2 g$, as is in fact the case, and

moreover the values give $x + 2g = 180^\circ$. Also each of the other angles is found to be $= 60^\circ$; that is, we have $\angle B_1BB_2 = 60^\circ$, or the whole angle at B_1 being $= 120^\circ$, the sum of the remaining angles B_1BB_2 and $BB_1\Theta$ is $= 60^\circ$.

From the triangle $\Theta B_1\Theta'$ where the two sides and the included angle are $\beta, \beta, 120^\circ$, we find $\Theta\Theta' = 36^\circ$.

And from the triangle $\Theta B_1\Theta''$, where the two sides and the included angle are g, g , and $(120^\circ - 2\phi)2\phi$, we find $\Theta\Theta'' = 60^\circ$.

99. We thus arrive at the following Table:

			cos	sin
$A\Theta$	α	$31^\circ 43'$	$\sqrt{\frac{5 - \sqrt{5}}{10}}$	$\sqrt{\frac{5 + \sqrt{5}}{10}}$
$B\Theta$	β	$20^\circ 55'$	$\frac{\sqrt{5} + 1}{2\sqrt{3}}$	$\frac{\sqrt{10 - 2\sqrt{5}}}{2\sqrt{3}}$
AB	γ	$37^\circ 22'$	$\sqrt{\frac{5 + \sqrt{5}}{15}}$	$\sqrt{\frac{10 - 2\sqrt{5}}{15}}$
(BB)	x	$70^\circ 32'$	$\frac{1}{3}$	$\frac{2\sqrt{2}}{3}$
(BB)		$109^\circ 28'$	$-\frac{1}{3}$	$\frac{2\sqrt{2}}{3}$
(BB)	g	$54^\circ 44'$	$\frac{1}{\sqrt{3}}$	$\frac{\sqrt{2}}{\sqrt{3}}$
BBB	θ	$37^\circ 46'$		
BBB		$54^\circ 44'$		
ΘBB	ϕ			
2β		$41^\circ 50'$	$\sqrt{\frac{5 - 2\sqrt{5}}{15}}$	
2γ		$74^\circ 44'$	$\frac{\sqrt{3}(\sqrt{5} - 1)}{4}$	
$\alpha - \beta$			$\frac{\sqrt{5 - \sqrt{6}}}{2}$	
$\Theta\Theta'$		36°	$\frac{\sqrt{5} + 1}{4}$	
$\Theta\Theta''$		60°	$\frac{1}{2}$	

where as above $\alpha + \beta + \gamma = 90^\circ$, and $x + 2g = 180^\circ$, $\theta + \phi = 60^\circ$.

100. We now construct three figures of the points A, B, Θ ; viz. these are stereographic projections, each showing the Northern hemisphere projected on the plane of the equator by lines drawn to the South Pole: hence, for any pair of opposite points on the equator, only the point in the Northern hemisphere is shown: but for a pair of opposite points on the equator the two points are each of them shown. In fig. 1 the North Pole is taken to be a point Θ ; in fig. 2 it is a point A ; and in fig. 3 it is a point B .

The position of any point on the sphere is determined by its N.P.D. and its longitude, say from the point E of the centre left-handedly measured from an arbitrary origin: then, in the three figures, the positions are as follows.

101. Fig. 1. Pole at Θ .

[Figure 1: Stereographic projection, Pole at Θ , showing positions of 12 A points, 20 B points and 3

	N.P.D.'s	Longitudes
$2A$	$\alpha = 31^\circ 43'$	$0^\circ, 180^\circ$
$2A$	$90^\circ - \alpha = 58\ 17$	$90, 270$
$2A$	$90^\circ + \alpha = 121\ 43$	$0^\circ, 180^\circ$
$2A$	$180^\circ - \alpha = 148\ 17$	$90, 270$
$4A$	90	$(0, 180) + \alpha = 31^\circ 43'$
$2B$	$\beta = 20^\circ 55'$	$90^\circ, 270^\circ$
$4B$	$g = 54\ 44$	$45, 135, 225, 315$
$2B$	$90^\circ - \beta = 69\ 5$	$0, 180$
$2B$	90	$(90, 270) + \beta = 20^\circ 55'$
$4B$	$90^\circ + \beta = 110\ 55$	$0, 180$
$4B$	$180^\circ - g = 125\ 16$	$45, 135, 225, 315$
$2B$	$180^\circ - \beta = 159\ 5$	$90, 270$
1Θ	0°	—
4Θ	36	$(90^\circ, 270^\circ) + \alpha = 31^\circ 43'$
4Θ	60	$(0, 180) + \beta = 20\ 55$
4Θ	72	$(90, 270) + \alpha = 31\ 43$
4Θ	90	$0, 90, 180, 270$
4Θ	108	$(90, 270) + \alpha = 31\ 43$
4Θ	120	$(0, 180) + \beta = 20\ 55$
4Θ	144	$(90, 270) + \alpha = 31\ 43$
1Θ	180	—

102. Fig. 2. Pole at A .

[Figure 2: Stereographic projection, Pole at A .]

	N.P.D.'s	Longitudes
A	0	—
$5A$	$2\alpha = 63^\circ 26'$	$0^\circ 72^\circ 144^\circ 216^\circ 288^\circ$
$5A$	$180^\circ - 2\alpha = 116\ 34$	36 108 180 252 324
A	180	—
$5B$	$\gamma = 37\ 22$	36 108 180 252 324
$5B$	$90^\circ - \alpha + \beta = 79\ 12$	36 108 180 252 324
$5B$	$90 + \alpha - \beta = 100\ 48$	72 144 216 288
$5B$	$180 - \gamma = 142\ 38$	72 144 216 288
5Θ	$\alpha = 31\ 43$	72 144 216 288
5Θ	$90^\circ - \alpha = 58\ 17$	36 108 180 252 324
10Θ	90	$(36\ 108\ 180\ 252\ 324) + 18^\circ$
5Θ	$90 + \alpha = 121\ 43$	72 144 216 288
5Θ	$180 - \alpha = 144\ 17$	36 108 180 252 324

103. Fig. 3. Pole at B .

[Figure 3: Stereographic projection, Pole at B .]

	N.P.D.'s	Longitudes
$3A$	$\gamma = 37^\circ 22'$	$30^\circ 150^\circ 270^\circ$
$3A$	$90^\circ - \alpha + \beta = 79\ 12$	90 210 330
$3A$	$90 + \alpha - \beta = 100\ 48$	90 210 330
$3A$	$180 - \gamma = 142\ 38$	90 210 330
B	0	—
$3B$	$2\beta = 41\ 50$	30 150 270
$6B$	$x = 70\ 32$	$(90\ 210\ 330) + \phi = 37^\circ 46'$
$6B$	$180^\circ - x = 109\ 28$	$(30\ 150\ 270) + \phi = 37^\circ 46'$
$3B$	$180 - 2\beta = 138\ 10$	90 210 330
B	180	—
3Θ	$\beta = 20\ 55$	90 210 330
6Θ	$g = 54\ 44$	$(90\ 210\ 330) + \phi = 22^\circ 14'$
Θ	$60\ 120\ 180^\circ 240^\circ 300^\circ$	—
6Θ	$90 + \beta = 110\ 55$	30 150 270
6Θ	$180 - g = 125\ 16$	30 150 270
3Θ	$180 - \beta = 159\ 5$	90 210 330

The groups of homographic transformations, resumed.

Art. Nos. 104 to 117.

104. The axes of rotation for the dodecahedron and the icosahedron are 15 axes each through a pair of opposite points Θ , 6 axes each through a pair of opposite points A , and 10 axes each through a pair of opposite points B ; or say 15 Θ -axes, 10 B -axes and 6 A -axes; the corresponding angles of

rotation are 180° , 72° and 120° so that (excluding in each case the original position or that of a rotation 0) we have in respect of each Θ -axis 1 position, in respect of each A -axis 4 positions, and in respect of each B -axis 2 positions; in all, including the original position,

$$1 + 15 + (6 \times 4) + (10 \times 2), \text{ that is, } = 60 \text{ positions,}$$

that is, a group of 60 rotations.

To find, in any one of the three forms, the group of homographic transformations, we can in each case obtain from the foregoing tables the values $\cos f$, $\cos g$, $\cos h$ of the cosine-inclination of an axis of rotation to the axes of coordinates, and thence calculate the values of

$$\lambda, \mu, \nu = \tan \frac{1}{2}\theta \cos f, \tan \frac{1}{2}\theta \cos g, \tan \frac{1}{2}\theta \cos h,$$

and thence the values of

$$A, B, C, D = -\nu - i, \lambda + i\mu, \lambda - i\mu, \nu - i;$$

viz. in the case of a Θ -axis, θ is $= 180^\circ$, (so that here $\tan \frac{1}{2}\theta = \infty$ or the values of A, B, C, D are $= -\nu, \lambda + i\mu, \lambda - i\mu, \nu$, that is, $-\cos h, \cos f + i \cos g, \cos f - i \cos g, \cos h$); in the case of a B -axis, the values are $\theta = 120^\circ, 240^\circ$, and therefore $\tan \frac{1}{2}\theta = \pm\sqrt{3}$; and in the case of an A -axis, they are $\theta = 72^\circ, 144^\circ, 216^\circ, 288^\circ$, and therefore

$$\tan \frac{1}{2}\theta = \pm \frac{1}{\sqrt{5}} \sqrt{10 + 2\sqrt{5}}, \pm \frac{1}{\sqrt{5}} \sqrt{10 - 2\sqrt{5}}.$$

105. The Θ -form was first given in my paper of 1879, but in obtaining it I used the results given in the paper of 1877. As regards the identification with the substitution-symbols, since there is nothing to distinguish inter se the letters a, b, c, d, e , any transformation A, B, C, D of the fifth order might have been taken for $abcde$, But No. 37 of the group having been taken for this substitution $abcde$, I do not recall in what manner I found that, consistently herewith, the transformation No. 2 ($-1, 0, 0, 1$, that is, s into $-s$) of the second order could be taken for $ab.cd$. But there is no sub-group of an order divisible by 5; and hence, these two transformations being identified with the two substitutions, the other transformations correspond each of them to a determinate substitution.

106. *Homographic Transformations.* The group of 60. Pole at Θ .

No.	$(Ax + B)$		$\div (Cx + D)$	substitution
1	1		1	1

No.	$(Ax + B)$		$\div (Cx + D)$	substitution
2	-1		1	$ab.cd$
3	i		1	$ac.bd$
4	$-i$		1	$ad.bc$
5	$-3 + \sqrt{5} + i(1 - \sqrt{5})$		2	$ad.be$
6	$-3 + \sqrt{5} + i(-1 + \sqrt{5})$		2	$ae.bd$
7	$3 - \sqrt{5} + i(-1 + \sqrt{5})$		2	$ad.ce$
8	$3 - \sqrt{5} + i(1 - \sqrt{5})$		2	$ae.cd$
9	$-1 - \sqrt{5} + i(1 - \sqrt{5})$		2	$ab.de$
10	$-1 - \sqrt{5} + i(-1 + \sqrt{5})$		2	$bc.de$
11	$1 + \sqrt{5} + i(-1 + \sqrt{5})$		2	$ab.ce$
12	$1 + \sqrt{5} + i(1 - \sqrt{5})$		2	$ac.bd$
13	$-1 - \sqrt{5} + i(-3 - \sqrt{5})$		2	$bd.ce$
14	$-1 - \sqrt{5} + i(3 + \sqrt{5})$		2	$ae.be$
15	$1 + \sqrt{5} + i(3 + \sqrt{5})$		2	$bc.de$
16	$1 + \sqrt{5} + i(-3 - \sqrt{5})$		2	$ac.de$
17	i		1	abc
18	$-i$		1	acb
19	1		$-i$	ade
20	-1		$-i$	aed
21	i		1	abd
22	$-i$		1	adb
23	1		i	bcd
24	-1		i	bdc
25	$-1 - \sqrt{5} + i(-3 - \sqrt{5})$		2	abd
26	$1 + \sqrt{5} + i(3 + \sqrt{5})$		2	ace
27	$1 + \sqrt{5} + i(-3 - \sqrt{5})$		2	bce
28	$-1 - \sqrt{5} + i(3 + \sqrt{5})$		2	bed
29	$-3 + \sqrt{5} + i(1 - \sqrt{5})$		2	bde
30	$-3 + \sqrt{5} + i(-1 + \sqrt{5})$		2	bee
31	$3 - \sqrt{5} + i(-1 + \sqrt{5})$		2	aed
32	$3 - \sqrt{5} + i(1 - \sqrt{5})$		2	ade
33	$-1 - \sqrt{5} + i(3 + \sqrt{5})$		2	cde
34	$1 + \sqrt{5} + i(-3 - \sqrt{5})$		2	ced
35	$1 + \sqrt{5} + i(3 + \sqrt{5})$		-2	aeb
36	$-1 - \sqrt{5} + i(-3 - \sqrt{5})$		2	abe
37	$1 + \sqrt{5} + i(-3 - \sqrt{5})$		2	$abcde$
38	$-1 - \sqrt{5} + i(1 - \sqrt{5})$		2	$acebd$
39	$-1 - \sqrt{5} + i(-1 + \sqrt{5})$		2	$adbec$
40	$-1 - \sqrt{5} + i(3 + \sqrt{5})$		2	$aedcb$
41	$1 + \sqrt{5} + i(3 + \sqrt{5})$		2	$adceb$
42	$1 + \sqrt{5} + i(-1 + \sqrt{5})$		2	$acbde$
43	$1 + \sqrt{5} + i(1 - \sqrt{5})$		2	$aedbe$
44	$1 + \sqrt{5} + i(-3 - \sqrt{5})$		2	$abcad$
45	$-1 - \sqrt{5} + i(-1 + \sqrt{5})$		-2	$abced$

No.	$(Ax + B)$		$\div (Cx + D)$	substitution
46	$-3 + \sqrt{5} + i(-1 + \sqrt{5})$		-2	<i>abdce</i>
47	$3 - \sqrt{5} + i(-1 + \sqrt{5})$		-2	<i>accdb</i>
48	$1 + \sqrt{5} + i(-1 + \sqrt{5})$		2	<i>adcbe</i>
49	$1 + \sqrt{5} + i(1 - \sqrt{5})$		2	<i>aecbd</i>
50	$3 - \sqrt{5} + i(1 - \sqrt{5})$		2	<i>acdeb</i>
51	$-3 + \sqrt{5} + i(1 - \sqrt{5})$		2	<i>abccd</i>
52	$-1 - \sqrt{5} + i(1 - \sqrt{5})$		2	<i>adbcc</i>
53	$-3 + \sqrt{5} + i(1 - \sqrt{5})$		-2	<i>acbde</i>
54	$3 - \sqrt{5} + i(1 - \sqrt{5})$		2	<i>aebde</i>
55	$-3 + \sqrt{5} + i(-1 + \sqrt{5})$		2	<i>abccd</i>
56	$-1 - \sqrt{5} + i(-3 - \sqrt{5})$		2	<i>adecb</i>
57	$1 + \sqrt{5} + i(3 + \sqrt{5})$		2	<i>acdbc</i>
58	$3 - \sqrt{5} + i(1 - \sqrt{5})$		2	<i>abdec</i>
59	$-3 + \sqrt{5} + i(1 - \sqrt{5})$		2	<i>adcbe</i>
60	$-1 - \sqrt{5} + i(-1 + \sqrt{5})$		2	<i>aebcd</i>

107. Taking out of the foregoing group of 60 a group of 12 contained in it, viz. that corresponding to the positive substitutions of the four letters *abcd*, it is easy to see, that there is a transformation $(i, 0, 0, 1)$, that is, *s* into *is*, which can be taken for the substitution *abcd*, and also to complete thence the group of 24. And we have thus the following Table.

Groups of 12 and 24. Pole at Θ .

No.	$(Ax + B)$		$\div (Cx + D)$	substitution
1	1		1	1
2	-1		1	<i>ab.cd</i>
3	<i>i</i>		1	<i>ac.bd</i>
4	- <i>i</i>		1	<i>ad.bc</i>
5	<i>i</i>		1	<i>abc</i>
6	- <i>i</i>		1	<i>acb</i>
7	1		- <i>i</i>	<i>adc</i>
8	-1		- <i>i</i>	<i>abd</i>
9	<i>i</i>		1	<i>acd</i>
10	- <i>i</i>		1	<i>adb</i>
11	1		<i>i</i>	<i>bcd</i>
12	-1		<i>i</i>	<i>bdc</i>
13	<i>i</i>		-1	<i>adbc</i>
14	- <i>i</i>		-1	<i>acbd</i>
15	1		-1	<i>ab.cd</i>
16	-1		-1	<i>aedb</i>
17	1		1	<i>bd</i>
18	-1		1	<i>abcd</i>
19	<i>i</i>		- <i>i</i>	<i>bc</i>
20	- <i>i</i>		- <i>i</i>	<i>abde</i>
21	-1		1	<i>ae</i>
22	1		1	<i>adeb</i>

No.	$(Ax + B)$		$\div (Cx + D)$	substitution
23	$-i$		$-i$	ad
24	i		-1	—

108. The group of 60 was obtained in the A -form by Gordan in his paper. The passage from the Θ -form to the A -form is made as follows: let X, Y, Z be the coordinates of a point when the axes are as in the Θ -form, X_1, Y_1, Z_1 the coordinates of the same point when the axes are as in the A -form; we may write

$$X_1, Y_1, Z_1 = aX + bZ, Y, -bX + aZ,$$

where

$$a = \sqrt{\frac{5 - \sqrt{5}}{10}}, \quad b = \sqrt{\frac{5 + \sqrt{5}}{10}},$$

then, if the equations $X : Y : Z = L : M : N$ are those of an axis of rotation referred to the first set of coordinates and $X_1 : Y_1 : Z_1 = L_1 : M_1 : N_1$, those of the same axis referred to the second set of coordinates, we may write

$$L_1, M_1, N_1 = aL + bN, M, -bL + aN;$$

these values are such that

$$L_1^2 + M_1^2 + N_1^2 = L^2 + M^2 + N^2,$$

and hence, λ, μ, ν and λ_1, μ_1, ν_1 being the rotations, we have

$$L, M, N = \frac{1}{\theta}\lambda, \frac{1}{\theta}\mu, \frac{1}{\theta}\nu; \quad L_1, M_1, N_1 = \frac{1}{\theta}\lambda_1, \frac{1}{\theta}\mu_1, \frac{1}{\theta}\nu_1;$$

where θ has the same value in each set of equations.

From the equations

$$A : B : C : D = -\nu - i : \lambda + i\mu : \lambda - i\mu : \nu - i,$$

we may write

$$B + C : B - C : D - A : D + A = \lambda : i\mu : \nu : -i, = L : iM : N : -i\theta,$$

and similarly

$$B_1 + C_1 : B_1 - C_1 : D_1 - A_1 : D_1 + A_1 = L_1 : iM_1 : N_1 : -i\theta.$$

Hence we may write

$$B_1 + C_1 = a(B + C) + b(D - A),$$

$$\begin{aligned}
B_1 - C_1 &= B - C, \\
D_1 - A_1 &= -b(B + C) + a(D - A), \\
D_1 + A_1 &= D + A;
\end{aligned}$$

or say,

$$\begin{aligned}
A_1 &= -a(B + C) - b(D - A) + (D + A), \\
B_1 &= a(B + C) + b(D - A) + (B - C), \\
C_1 &= a(B + C) + b(D - A) - (B - C), \\
D_1 &= -a(B + C) - b(D - A) + (D + A);
\end{aligned}$$

which are the values for a transformation (A_1, B_1, C_1, D_1) in the A -form: of course, as only the ratios are material, the values may be multiplied by any common factor.

109. The results are exhibited in terms of ϵ , an imaginary fifth root of unity: taking $\epsilon = \cos 72^\circ + i \sin 72^\circ$, we have

$$\begin{aligned}
\epsilon + \epsilon^4 &= \frac{\sqrt{5} - 1}{2}, & \epsilon^2 + \epsilon^3 &= -\frac{\sqrt{5} + 1}{2}, \\
a &= \sqrt{\frac{5 + \sqrt{5}}{10}} = \pm i \frac{\sqrt{5 + \sqrt{5}}}{\sqrt{10}}, & b &= \sqrt{\frac{5 - \sqrt{5}}{10}} = \pm \frac{\sqrt{5 - \sqrt{5}}}{\sqrt{10}},
\end{aligned}$$

where the upper signs belong to ϵ , ϵ^4 and the lower to ϵ^2 , ϵ^3 . It may be remarked that

$$\frac{1}{a} = \sqrt{\frac{5 + \sqrt{5}}{2}}, \quad \frac{1}{b} = \sqrt{\frac{5 - \sqrt{5}}{2}}, \quad \frac{a}{b} = \sqrt{\frac{\sqrt{5} + 1}{\sqrt{5} - 1}}.$$

For instance, we have in the Θ -group $(A, B, C, D) = (-1, 0, 0, 1)$; $ab.cd$: and thence in the A -group $A_1, B_1, C_1, D_1 = (-2b, 2a, 2a, 2b)$; $ab.cd$: or say this is

$$(-1, a/b, a/b, 1) = (-1, \epsilon + \epsilon^4, \epsilon + \epsilon^4, 1);$$

which in the Table is given as $(-\epsilon^*, \epsilon^2 + \epsilon^3, \epsilon^2 + \epsilon^3, \epsilon^*)$; $ab.cd$.

By effecting the passage to the A -group in this manner, we of course obtain the proper substitution corresponding to each transformation: but I found it easier starting from two transformations and the corresponding substitutions, to obtain by thence successive compositions the entire group.

110. *Homographic Transformations.* The group of 60. Pole at A .

No.	Θ No.	$(As + B)$		$\div (Cs + D)$		substitution
1	1	1	0	0	1	1
2	4	-1	0	0	1	$ad.bc$
3	13	$-\epsilon^4$	0	0	1	$ac.be$
4	9	$-\epsilon^4$	0	0	1	$ab.de$
5	10	$-\epsilon^3$	0	0	1	$bd.ce$
6	14	$-\epsilon$	0	0	1	$ae.be$
7	6	$\epsilon + \epsilon^4$	ϵ^2	ϵ^3	$-(\epsilon + \epsilon^4)$	$ae.bd$
8	5	$\epsilon + \epsilon^4$	ϵ^3	ϵ^2	$-(\epsilon + \epsilon^4)$	$ad.be$
9	15	$\epsilon + \epsilon^4$	ϵ^4	ϵ	$-(\epsilon + \epsilon^4)$	$bc.de$
10	3	$\epsilon + \epsilon^4$	ϵ	ϵ^4	$-(\epsilon + \epsilon^4)$	$ac.de$
11	12	$\epsilon + \epsilon^4$	ϵ^3	ϵ^2	$-(\epsilon + \epsilon^4)$	$ac.bd$
12	16	-1	$\epsilon^4 + \epsilon^3$	$\epsilon^2 + \epsilon$	1	$ae.bd$
13	11	$-\epsilon$	$\epsilon^3 + 1$	$1 + \epsilon^2$	ϵ	$ab.ce$
14	7	$-\epsilon^2$	$1 + \epsilon^4$	$\epsilon + \epsilon^4$	ϵ^2	$bc.cd$
15	2	$-\epsilon^3$	$\epsilon^3 + \epsilon$	$\epsilon^3 + \epsilon^4$	ϵ^3	$ab.ce$
16	8	$-\epsilon^4$	$\epsilon^2 + \epsilon$	$\epsilon^3 + \epsilon^4$	ϵ^4	$ad.ce$
17	21	$\epsilon^2 + 1$	ϵ	1	$-(\epsilon + \epsilon^2)$	adb
18	36	$\epsilon^4 + 1$	ϵ^2	ϵ^4	$-(\epsilon + \epsilon^3)$	aeb
19	30	$\epsilon^2 + 1$	ϵ^4	ϵ^3	$-(\epsilon^2 + \epsilon^4)$	bee
20	34	$\epsilon^2 + 1$	ϵ^3	ϵ^2	$-(\epsilon + \epsilon^2)$	cde
21	19	$\epsilon^2 + 1$	1	ϵ	$-(\epsilon + \epsilon^2)$	adc
22	33	$\epsilon + \epsilon^4$	1	ϵ^2	$-(\epsilon + \epsilon^2)$	cde
23	20	$\epsilon + \epsilon^4$	ϵ^3	ϵ^4	$-(\epsilon + \epsilon^3)$	acd
24	22	$\epsilon + \epsilon^4$	ϵ^4	ϵ	$-(\epsilon + \epsilon^2)$	abd
25	36	$\epsilon + \epsilon^4$	1	ϵ^4	$-(\epsilon + \epsilon^3)$	abe
26	29	$\epsilon + \epsilon^4$	ϵ	ϵ	$-(\epsilon + \epsilon^3)$	bee
27	31	$-\epsilon$	$\epsilon^2 + \epsilon^4$	$\epsilon^2 + \epsilon^4$	1	aed
28	17	$-\epsilon^2$	$\epsilon^2 + \epsilon$	$\epsilon^2 + \epsilon$	ϵ	abc
29	27	$-\epsilon^2$	$\epsilon^3 + \epsilon^4$	$\epsilon^3 + \epsilon^4$	ϵ^2	bcd
30	26	$-\epsilon^4$	$\epsilon^3 + \epsilon^4$	$\epsilon^3 + \epsilon^4$	ϵ^3	aec
31	23	-1	$\epsilon^2 + \epsilon^4$	$\epsilon^3 + \epsilon^4$	1	bcd
32	24	$-\epsilon^4$	1	ϵ^2	ϵ^2	bdc
33	32	-1	$\epsilon^2 + \epsilon^3$	$\epsilon^4 + \epsilon$	ϵ^4	ade
34	18	$-\epsilon$	$\epsilon + \epsilon^4$	$\epsilon^2 + \epsilon^3$	ϵ^2	aeb
35	28	$-\epsilon^2$	$\epsilon + \epsilon^4$	$\epsilon^2 + \epsilon^3$	ϵ^4	bde
36	25	$-\epsilon^3$	$\epsilon^3 + \epsilon^4$	$\epsilon^3 + \epsilon^4$	ϵ^4	ace
37	44	ϵ	1	1	1	$abccd$
38	43	ϵ^4	1	1	1	$aedbc$
39	42	ϵ^3	1	1	1	$acbde$
40	41	ϵ^2	1	1	1	$adceb$
41	38	$\epsilon^3 + \epsilon^4$	1	1	$-(\epsilon + \epsilon^3)$	$acebd$
42	46	$\epsilon^3 + \epsilon^4$	ϵ	ϵ^4	$-(\epsilon + \epsilon^3)$	$abdcc$
43	58	$\epsilon^2 + \epsilon^4$	ϵ^2	ϵ^3	$-(\epsilon + \epsilon^3)$	$adcbc$
44	55	$\epsilon^2 + \epsilon^4$	ϵ^3	ϵ^2	$-(\epsilon + \epsilon^3)$	$adecb$
45	50	$\epsilon^3 + \epsilon^4$	ϵ^4	ϵ	$-(\epsilon + \epsilon^3)$	$acdeb$
46	51	$\epsilon^2 + \epsilon^4$	ϵ^2	1	$-(\epsilon + \epsilon^3)$	$abedc$
47	39	$1 + \epsilon^2$	ϵ^2	ϵ^4	$-(\epsilon + \epsilon^3)$	$adbcc$
48	47	$1 + \epsilon^2$	1	ϵ^3	$-(\epsilon + \epsilon^3)$	$accdb$
49	59	$1 + \epsilon^2$	ϵ	ϵ^2	$-(\epsilon + \epsilon^3)$	$acbcd$
50	54	$1 + \epsilon^2$	ϵ^3	ϵ	$-(\epsilon + \epsilon^3)$	$abced$
51	56	$-\epsilon^2$	$1 + \epsilon^4$	$\epsilon^2 + \epsilon^3$	ϵ^3	$abdec$
52	49	$-\epsilon^4$	$\epsilon + \epsilon^3$	$\epsilon^2 + \epsilon^4$	ϵ	$abdcc$
53	37	$-\epsilon^3$	$\epsilon^4 + \epsilon$	$\epsilon^2 + \epsilon^3$	ϵ^4	$aebcd$

No.	Θ No.	$(As + B)$		$\div (Cs + D)$		substitution
54	45	-1	$\epsilon + \epsilon^2$	$\epsilon^3 + \epsilon^4$	ϵ^2	<i>adebc</i>
55	57	$-\epsilon$	$\epsilon + \epsilon^2$	$\epsilon^3 + \epsilon^4$	ϵ^4	<i>abdec</i>
56	48	$-\epsilon^3$	$\epsilon^4 + \epsilon$	$\epsilon^2 + \epsilon^3$	1	<i>adcbc</i>
57	60	$-\epsilon^2$	$\epsilon + \epsilon^3$	$\epsilon^2 + \epsilon^4$	ϵ	<i>acedb</i>
58	53	$\epsilon^3 + 1$	$\epsilon^2 + \epsilon^4$	ϵ^2	ϵ^3	<i>aebdc</i>
59	52	$-1 + \epsilon^4$	$\epsilon^2 + \epsilon^3$	ϵ^2	ϵ^3	<i>adbcc</i>
60	40	$-\epsilon^2$	$\epsilon + \epsilon^4$	ϵ^2	ϵ^2	<i>aedcb</i>

[PAPER 745 CONTINUES BEYOND P. 201 WITH ART. 111 SELECTING THE POSITIVE SUBSTITUTIONS *abcd* AND COMPLETING THE GROUP OF 24, POLE AT *A*, ETC.]